

Representation Analysis of Dermatological Images Using Self-Supervised Visual Features

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1. Introduction

This report presents an objective evaluation of pretrained self-supervised visual representations for dermatological image understanding. We investigate whether models like DINOv2 (vision-only self-supervised), CLIP (contrastive vision-language pretraining), and ResNet50 (supervised ImageNet) encode relevant medical concepts in their learned embedding spaces without domain-specific clinical fine-tuning. Using post-hoc clustering, nearest-neighbor retrieval, and centroid distance analysis, we evaluate how well these representations map to human-defined dermatological disease categories.

2. Key Findings

2.1 Fixed 23-Class Clustering Results

- **Best Representation by Silhouette Score:** DINOv2 using **kmeans** (Score: **0.1609**), suggesting high visual cluster compactness.
- **Best Representation by ARI:** CLIP using **kmeans** (Score: **0.1001**), suggesting alignment with clinical labels.
- **Best Representation by NMI:** CLIP using **kmeans** (Score: **0.1829**).
- **Best Representation by Purity:** CLIP using **kmeans** (Score: **0.2674**).
- **Worst Performing Representation:** RESNET50 using **kmeans** (Score: **0.0499**), suggesting poor generalization of ImageNet features to dermatological domains.
- **Best Clustering Algorithm:** **kmeans**.
- **Worst Clustering Algorithm:** **agglomerative**.

2.2 Variable HDBSCAN Clustering Results

- **Best Nominal Silhouette Score (HDBSCAN):** RESNET50 using **hdbscan** (Score: **0.2581**). However, this value is not directly comparable to the 23-class clustering methods, as it represents a heavily under-clustered space where the algorithm identified only 3 to 5 clusters and marked a large portion of the dataset as noise.
- **HDBSCAN Evaluation:** HDBSCAN clustering failed to align with ground-truth labels (mean ARI ~0.0). It suffered from significant under-clustering, finding only 3 to 5 clusters instead of the required 23, classifying a high proportion of the dataset as noise (up to 43.8% for ResNet50).

2.3 Overall Recommendation

We recommend **DINOv2** as the primary representation depending on the downstream application. CLIP is recommended for semantic or label-aligned categorization tasks, whereas DINOv2 is recommended for visual similarity retrieval and spatial compactness.

3. Dataset & Preprocessing

We utilize the DermNet dataset, containing clinical images of skin lesions across various disease categories. Clinical datasets often suffer from near-duplicate images due to multiple shots of the same patient lesion under slightly different lighting or angles. Such duplicates inflate evaluations and bias unsupervised cluster densities.

- **Total Raw Images:** 19559
- **Number of ground-truth classes:** 23
- **Deduplication Method:** SHA-256 hash comparison of image binaries
- **Near-Duplicates Removed:** 1257 images
- **Remaining Images for Analysis:** 18302
- **Largest Class:** Psoriasis pictures Lichen Planus and related diseases (1757)
- **Smallest Class:** Urticaria Hives (265)

Preprocessing Impact

Duplicate removal ensures that clustering algorithms are not biased towards highly dense subregions consisting of identical lesion files, which would artificially skew the silhouette width and purity. It also guarantees that nearest-neighbor retrieval is evaluated on distinct clinical images, representing a true out-of-sample similarity challenge rather than exact matching.

4. Methodology

Our evaluation pipeline operates as follows:

1. **Feature Extraction:** Images are resized, normalized according to pretraining specifications, and passed through pretrained encoders (DINOv2, CLIP, ResNet50) to yield high-dimensional visual descriptors.

- 2. **Dimensionality Reduction:** PCA (50 principal components) is used to standardise variance and remove noise before clustering. UMAP (2 components, cosine metric) is applied strictly for visual analysis.
- 3. **Post-hoc Clustering:** K-Means, Agglomerative (ward linkage), and HDBSCAN are run on the PCA space. Metrics are calculated both with and without ground-truth alignment.
- 4. **Geometric Profiling:** Pairwise cosine distances between class centroids are calculated to inspect the visual topology.
- 5. **Nearest-Neighbor Retrieval:** Query-response retrieval is performed on the raw embeddings to assess local neighborhood structure.

5. Feature Encoders & Computational Performance

Model	Architecture	Pretraining Objective	Parameters	Dim	Preprocessing Resolution	Feature Extraction Time
DINOv2	ViT-B/14	Self-supervised (DINO + iBOT)	~86 M	768	224x224	~670 s (~27 img/s)
ResNet50	CNN	Supervised (ImageNet-1K)	~25 M	2048	224x224	~135 s (~135 img/s)
CLIP	ViT-B/32	Contrastive Language-Image	~86 M (Img En)	512	224x224	~230 s (~80 img/s)

Note: Extraction runtimes were profiled on Apple Silicon (MPS). ViT architectures (DINOv2 and CLIP) incur higher computational overhead due to self-attention patches compared to ResNet50. However, DINOv2's smaller patch size (14 vs 32 in CLIP) renders it substantially more expensive computational-wise, taking nearly 3x longer than CLIP.

6. Dimensionality Reduction & Visualization

PCA reduces the high-dimensional features to 50 dimensions. UMAP is then used to project the data into 2D space. The following figures represent the UMAP layouts colored by disease category.

DINOv2

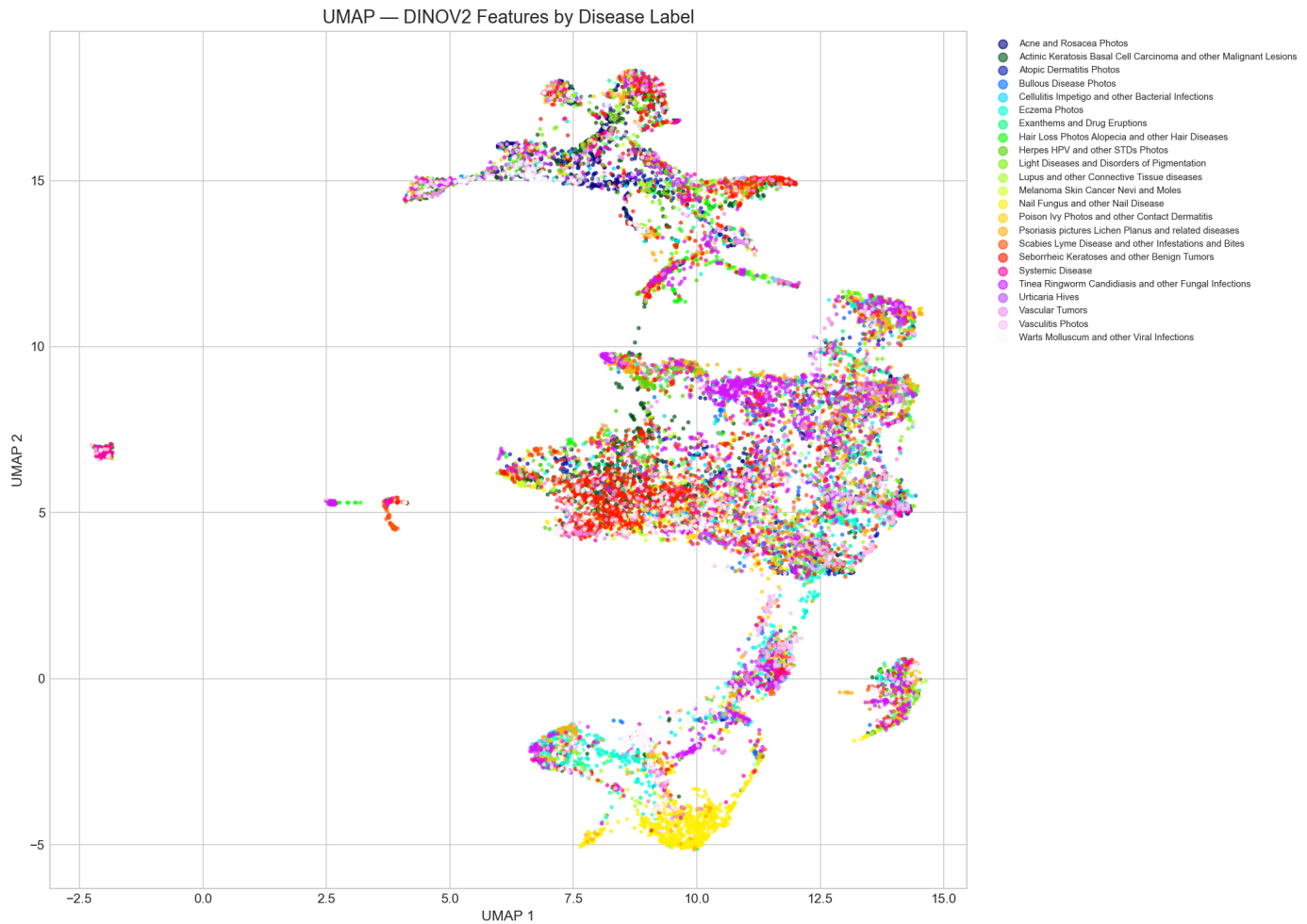


Figure DINOv2-UMAP: 2D UMAP projection of DINOv2 embedding space colored by ground-truth labels. Observe whether distinct colors form isolated clusters (semantic separation) or overlap extensively (high visual confusion).

RESNET50

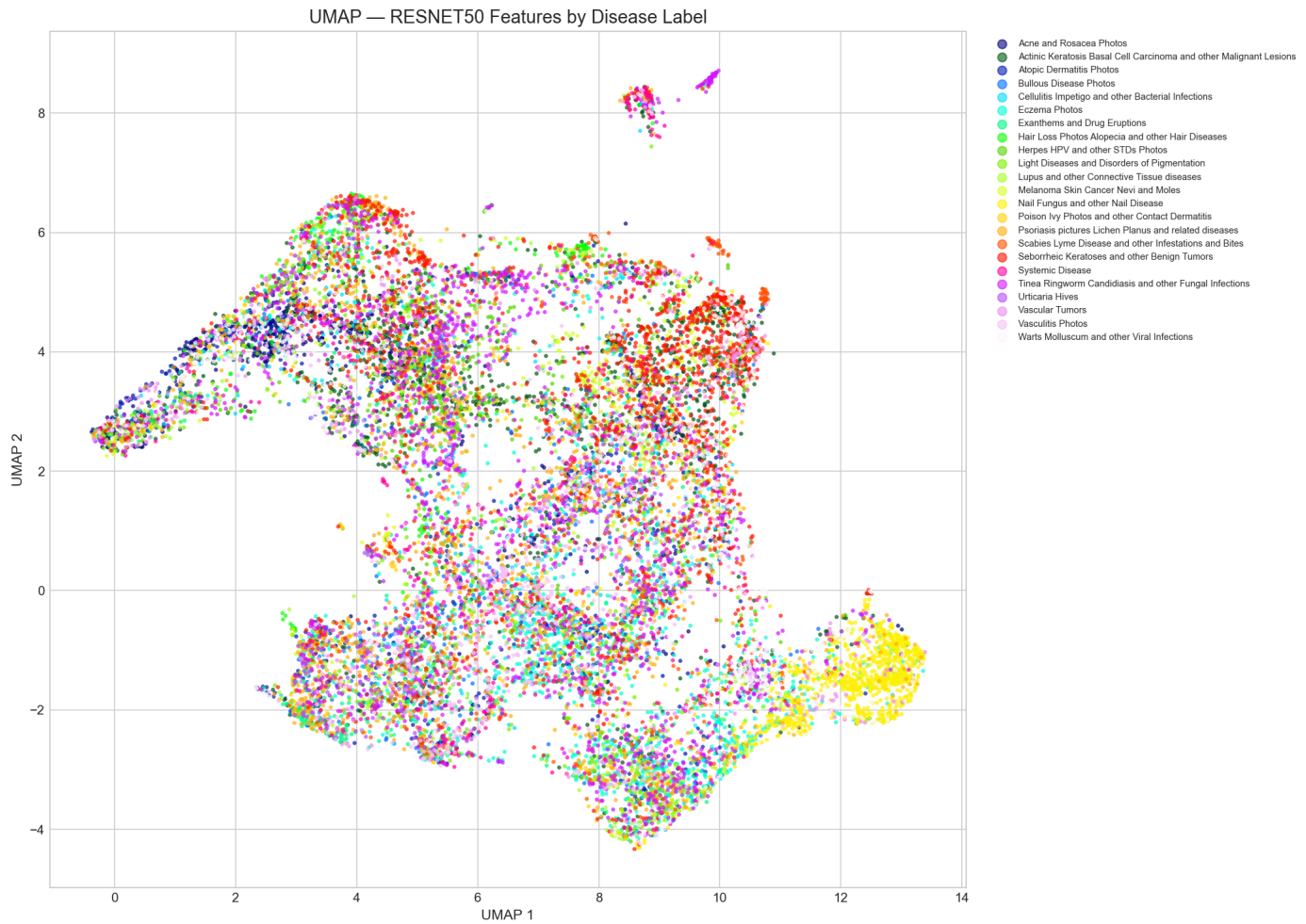


Figure RESNET50-UMAP: 2D UMAP projection of RESNET50 embedding space colored by ground-truth labels. Observe whether distinct colors form isolated clusters (semantic separation) or overlap extensively (high visual confusion).

CLIP

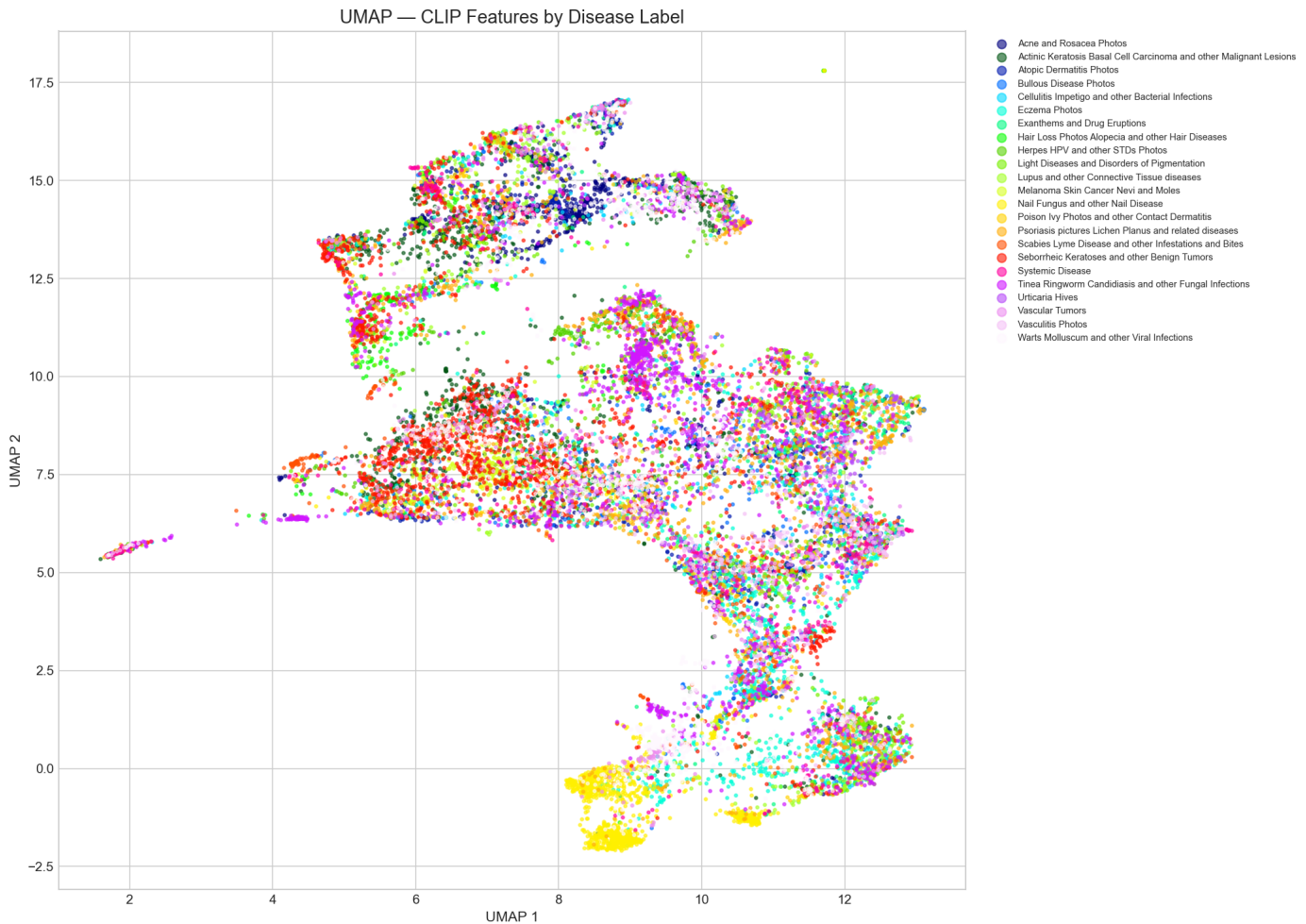


Figure CLIP-UMAP: 2D UMAP projection of CLIP embedding space colored by ground-truth labels. Observe whether distinct colors form isolated clusters (semantic separation) or overlap extensively (high visual confusion).

7. Clustering as Post-hoc Evaluation

We treat clustering performance as an indicator of embedding quality. A model whose representations natively group according to human disease categories will display higher ARI and Purity.

7.1 Clustering Metrics

representation	clustering_method	n_clusters	silhouette	davies_bouldin	calinski_harabasz	ari	nmi	purity
dinov2	kmeans	23	0.1609	1.8508	1083.68	0.0922	0.1796	0.2626
dinov2	agglomerative	23	0.128	2.0177	923.21	0.0879	0.18	0.2566
dinov2	hdbscan	5	0.115	1.1211	227.55	0.0002	0.0199	0.099
resnet50	kmeans	23	0.0451	2.3256	552.1	0.0499	0.1279	0.2151
resnet50	agglomerative	23	-0.0033	2.5688	425.76	0.0531	0.1285	0.2081
resnet50	hdbscan	3	0.2581	1.0273	127.41	-0.0002	0.0071	0.0941
clip	kmeans	23	0.0965	2.2947	671.38	0.1001	0.1829	0.2674
clip	agglomerative	23	0.0692	2.6454	563.39	0.0889	0.1788	0.2586
clip	hdbscan	3	0.217	1.0743	262.14	-0.0004	0.0115	0.0963

7.2 Interpretation of Absolute Metric Magnitudes

An analysis of the absolute values indicates that unsupervised clustering on raw embeddings remains a highly challenging task. The best-performing model (CLIP) achieves an Adjusted Rand Index (ARI) of 0.1001 and a Normalized Mutual Information (NMI) of 0.1829. These scores suggest only modest agreement with ground-truth clinical labels. This is expected because skin diseases represent fine-grained categories that are often defined by subtle histopathological or clinical criteria rather than coarse image-level visual distinctions. High intra-class variance (caused by differences in lighting, camera angles, and lesion stages) combined with high inter-class visual similarity (different diseases presenting with similar redness or scaling) limits the performance of standard clustering algorithms. However, despite low absolute clustering metrics, the embedding space exhibits meaningful visual organization, as demonstrated by the nearest-neighbor retrieval top-1 accuracy (up to 63.91% for DINOv2).

7.3 Analysis of Clustering Performance

The clustering results reveal that **CLIP** (using **kmeans**) achieves the highest Adjusted Rand Index (ARI: **0.1001**) and Purity (Purity: **0.2674**), suggesting that contrastive image-text training (CLIP) aligns the visual representation space more closely with human-defined medical taxonomies. For full 23-class clustering, **DINOv2** achieves the highest Silhouette score (Silhouette: **0.1609** using **kmeans**), indicating that its vision-only features yield visually tighter, more coherent partition boundaries in the spatial structure. While HDBSCAN runs report high nominal Silhouette scores (e.g., **0.2581** for **RESNET50**), this is an artifact of severe under-clustering where only 3 dense clusters are kept and the rest is flagged as noise. **ResNet50** exhibits the weakest performance for label classification (K-Means ARI: **0.0499**), confirming that ImageNet-1K supervised features suffer from heavy object bias and fail to generalize effectively to fine-grained clinical skin lesions without fine-tuning.

8. Representation Space Geometry

We compute class centroids in embedding space and calculate pairwise distances to inspect class separation.

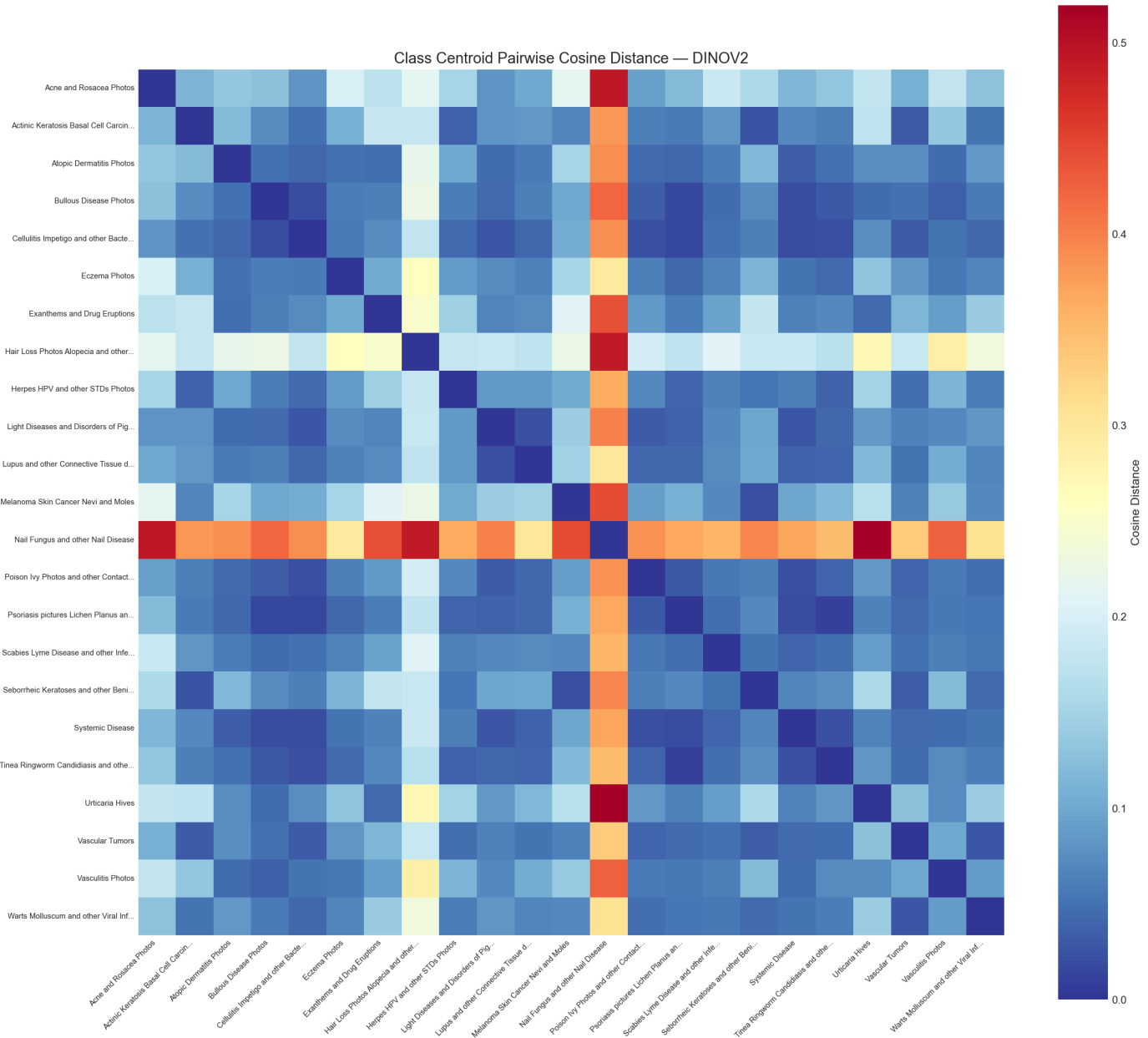


Figure DINOv2-Centroids: Pairwise cosine distance matrix of class centroids in DINOv2 space. Red blocks indicate highly proximate classes (potential confusion), while blue blocks indicate well-separated categories.

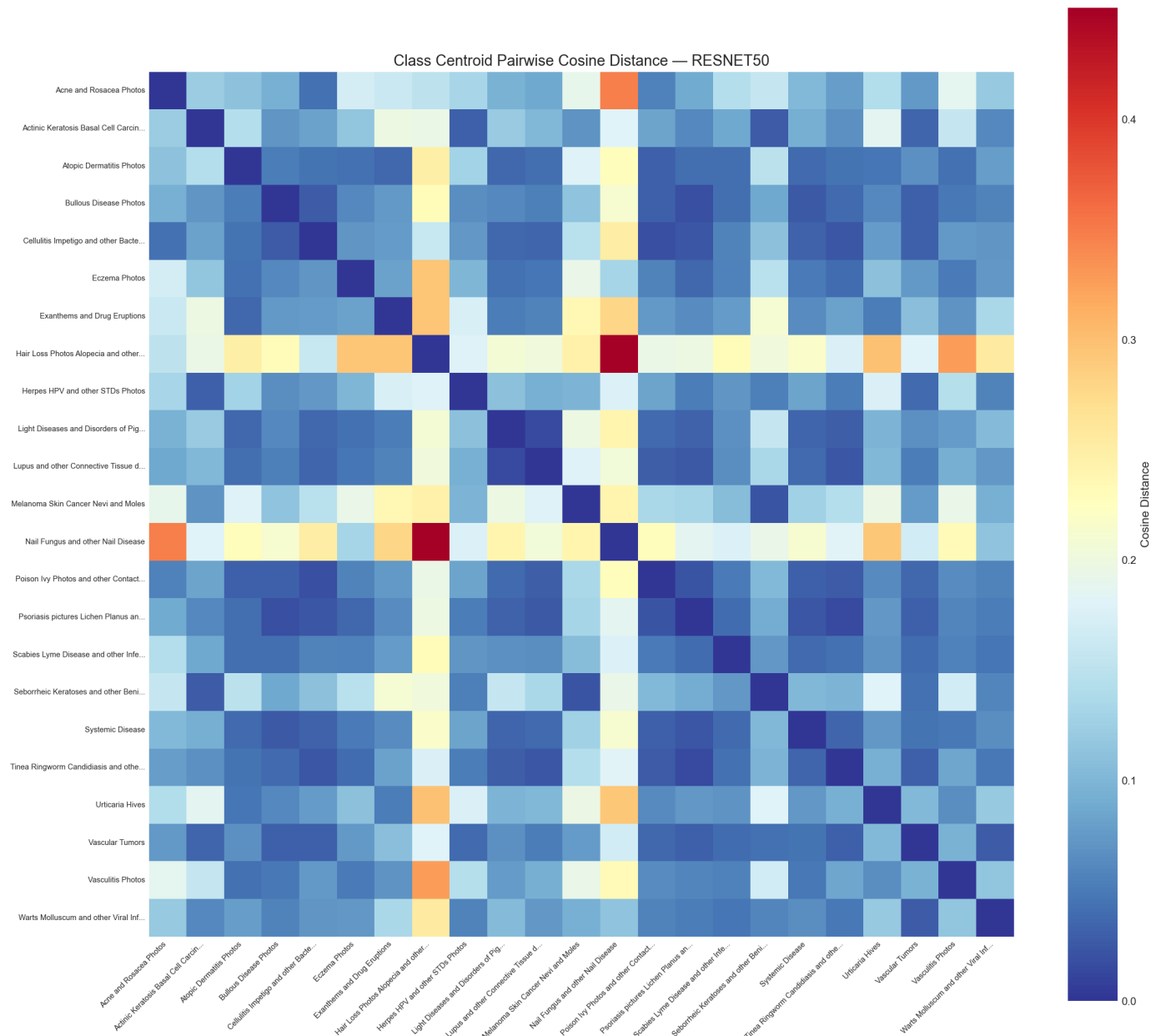


Figure RESNET50-Centroids: Pairwise cosine distance matrix of class centroids in RESNET50 space. Red blocks indicate highly proximate classes (potential confusion), while blue blocks indicate well-separated categories.

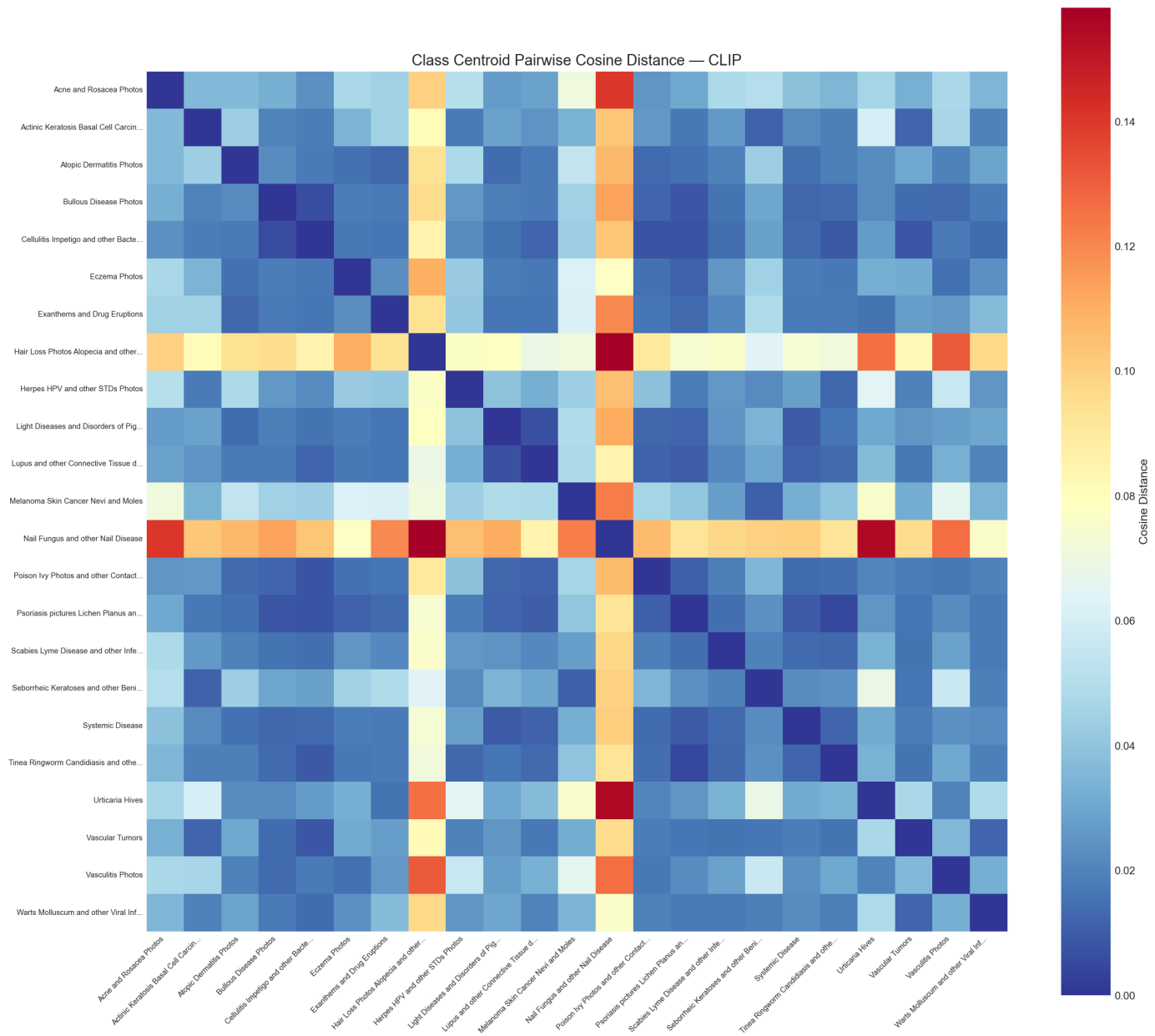


Figure CLIP-Centroids: Pairwise cosine distance matrix of class centroids in CLIP space. Red blocks indicate highly proximate classes (potential confusion), while blue blocks indicate well-separated categories.

Centroid Distances and Compactness

representation	disease_class	n_samples	intra_class_distance	intra_class_std	nearest_neighbor_class	centroid_distance_to_nearest
dinov2	Acne and Rosacea Photos	862	0.2344	0.1424	Light Diseases and Disorders of Pigmentation	0.0796
dinov2	Actinic Keratosis Basal Cell Carcinoma and other Malignant Lesions	1393	0.2861	0.1681	Seborrheic Keratoses and other Benign Tumors	0.0251
dinov2	Atopic Dermatitis Photos	603	0.3394	0.164	Systemic Disease	0.0342
dinov2	Bullous Disease Photos	544	0.2848	0.1797	Psoriasis pictures Lichen Planus and related diseases	0.0157

representation	disease_class	n_samples	intra_class_distance	intra_class_std	nearest_neighbor_class	centroid_distance_to_nearest
dinov2	Cellulitis Impetigo and other Bacterial Infections	351	0.3012	0.1443	Psoriasis pictures Lichen Planus and related diseases	0.0156
dinov2	Eczema Photos	1422	0.2836	0.1248	Psoriasis pictures Lichen Planus and related diseases	0.0414
dinov2	Exanthems and Drug Eruptions	468	0.3411	0.1343	Urticaria Hives	0.0442
dinov2	Hair Loss Photos Alopecia and other Hair Diseases	277	0.3396	0.1624	Tinea Ringworm Candidiasis and other Fungal Infections	0.1688
dinov2	Herpes HPV and other STDs Photos	487	0.2768	0.1614	Tinea Ringworm Candidiasis and other Fungal Infections	0.0375
dinov2	Light Diseases and Disorders of Pigmentation	686	0.3144	0.1435	Lupus and other Connective Tissue diseases	0.0234
dinov2	Lupus and other Connective Tissue diseases	519	0.3672	0.1389	Light Diseases and Disorders of Pigmentation	0.0234
dinov2	Melanoma Skin Cancer Nevi and Moles	557	0.2617	0.1712	Seborrheic Keratoses and other Benign Tumors	0.0218
dinov2	Nail Fungus and other Nail Disease	1164	0.1925	0.1568	Eczema Photos	0.2954
dinov2	Poison Ivy Photos and other Contact Dermatitis	316	0.3183	0.168	Systemic Disease	0.0204
dinov2	Psoriasis pictures Lichen Planus and related diseases	1590	0.3223	0.1596	Tinea Ringworm Candidiasis and other Fungal Infections	0.0096
dinov2	Scabies Lyme Disease and other Infestations and Bites	522	0.3802	0.2261	Systemic Disease	0.0396
dinov2	Seborrheic Keratoses and other Benign Tumors	1685	0.2896	0.1741	Melanoma Skin Cancer Nevi and Moles	0.0218
dinov2	Systemic Disease	739	0.352	0.2033	Psoriasis pictures Lichen Planus and related diseases	0.0194

representation	disease_class	n_samples	intra_class_distance	intra_class_std	nearest_neighbor_class	centroid_distance_to_nearest
dinov2	Tinea Ringworm Candidiasis and other Fungal Infections	1554	0.324	0.1944	Psoriasis pictures Lichen Planus and related diseases	0.0096
dinov2	Urticaria Hives	261	0.216	0.1098	Exanthems and Drug Eruptions	0.0442
dinov2	Vascular Tumors	592	0.3365	0.1305	Warts Molluscum and other Viral Infections	0.027
dinov2	Vasculitis Photos	510	0.2302	0.1511	Bullous Disease Photos	0.0361
dinov2	Warts Molluscum and other Viral Infections	1200	0.2937	0.1387	Vascular Tumors	0.027

Across all classes, DINOv2 embeddings show an average intra-class distance of **0.2994** compared to **0.1328** for CLIP and **0.3377** for ResNet50. This indicates that DINOv2 generates tighter local neighborhoods, aligning with its higher Silhouette scores.

9. Disease Confusion Analysis

We identify pairs of diseases that are consistently mapped to adjacent regions in embedding space. Understanding these confusions helps discover shared visual characteristics.

Confused Disease Pairs and Hypothesized Visual Causes

Rank	Representation	Disease A	Disease B	Confusion Score	Visual Hypothesis / Biological Reason
1	dinov2	Psoriasis pictures Lichen Planus and rel...	Tinea Ringworm Candidiasis and other Fun...	1.3622	The embedding proximity may be influenced by similar visual morphology, specifically erythematous plaques displaying peripheral scale textures and annular boundaries.
2	dinov2	Cellulitis Impetigo and other Bacterial ...	Poison Ivy Photos and other Contact Derm...	1.3447	Both diseases frequently manifest as poorly defined erythematous patches, localized swelling, and oozing vesicles, presenting overlapping hue and shape descriptors.
3	dinov2	Bullous Disease Photos...	Systemic Disease...	1.3352	Both conditions can present with raw, eroded skin surfaces and widespread blistering, which may confuse features in the absence of localized context.
4	dinov2	Actinic Keratosis Basal Cell Carcinoma a...	Seborrheic Keratoses and other Benign Tu...	1.3197	Both represent hyperkeratotic, pigmented plaques on sun-damaged skin, showing visual overlap in crusty surface textures and brown-tan color distributions.
5	dinov2	Light Diseases and Disorders of Pigmenta...	Lupus and other Connective Tissue diseases...	1.3027	Connective tissue disorders like cutaneous lupus frequently present as photosensitive rashes, which closely resemble primary photodermatoses in location and erythema.
6	dinov2	Atopic Dermatitis Photos...	Systemic Disease...	1.3019	Generalized eczematous patches with diffuse dry scaling and excoriations lack clear margins, likely resulting in highly overlapping global spatial features.
7	dinov2	Light Diseases and Disorders of Pigmenta...	Systemic Disease...	1.2937	Photosensitive cutaneous involvement in systemic autoimmune diseases is concentrated on sun-exposed regions, mimicking photodermatoses.
8	dinov2	Cellulitis Impetigo and other Bacterial ...	Vascular Tumors...	1.2916	Both categories present as raised, localized, red-to-purple nodules, likely creating highly overlapping color and shape statistics.
9	dinov2	Systemic Disease...	Vasculitis Photos...	1.2819	Palpable purpura and necrotic ulcers in vasculitis represent cutaneous findings that visually mirror other eruptive systemic conditions.
10	dinov2	Bullous Disease Photos...	Cellulitis Impetigo and other Bacterial ...	1.2808	Ruptured bullae result in crusted, erythematous zones that visually mimic localized impetigo crusts or cellulitis.

True Disease Class	Cluster																								
	C0	C1	C11	C12	C5	C6	C6	C7	C8	C10	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	
Acne and Rosacea Photos	282	13	0	5	194	1	22	0	15	4	5	16	0	96	101	9	14	4	39	28	12	2	0		
Actinic Keratosis Basal Cell C...	132	92	36	79	83	5	29	21	190	30	13	203	1	53	45	22	226	36	44	20	0	10	23		
Atopic Dermatitis Photos	13	6	134	1	34	50	22	2	3	19	20	20	0	14	39	8	16	7	22	17	84	66	6		
Bullous Disease Photos	10	0	34	25	10	40	42	2	15	15	4	74	0	1	96	0	15	17	47	46	4	39	8		
Cellulitis Impetigo and other ...	37	16	30	4	29	24	13	2	22	15	2	40	0	9	36	1	2	1	26	12	8	16	6		
Eczema Photos	17	21	232	0	7	214	27	12	32	74	209	96	21	23	90	7	12	6	42	52	2	95	131		
Exanthems and Drug Eruptions	4	2	34	5	12	10	80	1	14	17	39	19	4	3	32	0	0	1	36	55	54	39	7		
Hair Loss Photos Alopecia and ...	15	19	0	0	14	0	0	138	10	0	0	0	2	13	16	15	9	0	2	19	4	1	0		
Herpes HPV and other STDs Photos	13	6	15	29	4	6	10	7	213	3	2	62	1	3	39	10	19	1	27	0	7	6	4		
Light Diseases and Disorders o...	75	28	79	11	69	2	43	1	18	134	3	19	2	38	24	11	7	3	32	32	9	46	0		
Lupus and other Connective Tis...	57	14	9	5	41	17	29	20	9	69	36	33	23	19	25	5	7	10	33	30	7	6	15		
Melanoma Skin Cancer Nevi and ...	17	20	6	7	19	5	5	21	6	1	0	159	1	6	11	32	170	0	25	30	4	1	11		
Nail Fungus and other Nail Dis...	4	1	23	0	0	45	3	8	2	20	4	34	923	1	12	7	5	2	2	3	0	1	64		
Poison Ivy Photos and other Co...	33	12	45	3	11	20	11	1	5	8	3	62	0	30	21	9	6	0	11	13	0	12	0		
Psoriasis pictures Lichen Plan...	49	53	65	56	24	85	114	73	143	57	84	157	27	39	166	13	21	31	109	121	15	58	30		
Scabies Lyme Disease and other...	1	1	42	0	8	28	24	7	31	34	1	60	0	3	26	86	43	1	50	9	9	35	23		
Seborrheic Keratoses and other...	57	113	51	4	66	44	16	108	78	13	7	319	3	69	94	35	451	26	61	27	9	18	16		
Systemic Disease	50	10	75	14	24	46	30	5	5	24	18	77	7	38	56	14	14	46	59	32	9	72	14		
Tinea Ringworm Candidiasis and...	48	6	43	60	50	132	49	59	187	77	51	121	2	10	131	84	12	5	244	75	18	25	65		
Urticaria Hives	2	1	16	0	5	0	37	0	1	1	3	24	0	0	54	1	0	0	44	35	6	31	0		
Vascular Tumors	35	23	21	29	34	27	36	5	36	8	2	57	13	20	21	4	78	0	18	23	50	11	41		
Vasculitis Photos	11	0	96	3	0	30	5	0	1	18	16	62	0	1	65	0	12	8	10	13	10	143	6		
Warts Molluscum and other Vira...	138	4	23	51	75	41	19	2	20	43															

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The difference in performance between these models is rooted in their training objectives. DINOv2 utilizes a self-supervised visual objective (DINO self-distillation + iBOT masked image modeling) which forces the network to retain high-frequency spatial details, local textures, and boundary contours, making it ideal for content-based visual search. CLIP uses a multimodal contrastive objective, aligning global image embeddings with text representations. This global semantic pooling discards fine-grained local textures in favor of broad semantic concepts, explaining why it aligns better with high-level clinical categories but produces less compact visual clusters. Supervised ResNet50 is optimized to classify coarse classes (e.g. dog breeds, household objects) and actively discards features that do not contribute to that specific 1,000-class task, making it blind to the visual nuances of skin lesions.

11. Qualitative Visual Analysis

Nearest Neighbor Retrieval Performance

representation	disease_class	n_queries	top1_accuracy	top5_accuracy
dinov2	Acne and Rosacea Photos	10	0.8	0.9
dinov2	Actinic Keratosis Basal Cell Carcinoma and other Malignant Lesions	10	0.9	1
dinov2	Atopic Dermatitis Photos	10	0.6	0.7
dinov2	Bullous Disease Photos	10	0.7	0.7
dinov2	Cellulitis Impetigo and other Bacterial Infections	10	0.4	0.8
dinov2	Eczema Photos	10	0.7	0.8
dinov2	Exanthems and Drug Eruptions	10	0.5	0.5
dinov2	Hair Loss Photos Alopecia and other Hair Diseases	10	0.8	1
dinov2	Herpes HPV and other STDs Photos	10	0.8	0.8
dinov2	Light Diseases and Disorders of Pigmentation	10	0.7	0.9
dinov2	Lupus and other Connective Tissue diseases	10	0.4	0.8
dinov2	Melanoma Skin Cancer Nevi and Moles	10	0.6	0.9
dinov2	Nail Fungus and other Nail Disease	10	0.8	0.9
dinov2	Poison Ivy Photos and other Contact Dermatitis	10	0.2	0.5
dinov2	Psoriasis pictures Lichen Planus and related diseases	10	0.7	1
dinov2	Scabies Lyme Disease and other Infestations and Bites	10	0.5	0.5
dinov2	Seborrheic Keratoses and other Benign Tumors	10	0.8	0.8
dinov2	Systemic Disease	10	0.1	0.4
dinov2	Tinea Ringworm Candidiasis and other Fungal Infections	10	0.7	0.9
dinov2	Urticaria Hives	10	1	1
dinov2	Vascular Tumors	10	0.5	0.8
dinov2	Vasculitis Photos	10	0.8	0.9
dinov2	Warts Molluscum and other Viral Infections	10	0.7	0.9
resnet50	Acne and Rosacea Photos	10	0.5	0.8
resnet50	Actinic Keratosis Basal Cell Carcinoma and other Malignant Lesions	10	1	1
resnet50	Atopic Dermatitis Photos	10	0.6	0.6
resnet50	Bullous Disease Photos	10	0.7	0.7
resnet50	Cellulitis Impetigo and other Bacterial Infections	10	0.6	0.7
resnet50	Eczema Photos	10	0.5	0.8
resnet50	Exanthems and Drug Eruptions	10	0.6	0.6
resnet50	Hair Loss Photos Alopecia and other Hair Diseases	10	0.8	1
resnet50	Herpes HPV and other STDs Photos	10	0.5	0.7
resnet50	Light Diseases and Disorders of Pigmentation	10	0.6	0.7
resnet50	Lupus and other Connective Tissue diseases	10	0.4	0.7
resnet50	Melanoma Skin Cancer Nevi and Moles	10	0.5	0.7
resnet50	Nail Fungus and other Nail Disease	10	0.7	1
resnet50	Poison Ivy Photos and other Contact Dermatitis	10	0.2	0.2

representation	disease_class	n_queries	top1_accuracy	top5_accuracy
resnet50	Psoriasis pictures Lichen Planus and related diseases	10	0.4	0.7
resnet50	Scabies Lyme Disease and other Infestations and Bites	10	0.4	0.5
resnet50	Seborrheic Keratoses and other Benign Tumors	10	0.5	0.7
resnet50	Systemic Disease	10	0.1	0.4
resnet50	Tinea Ringworm Candidiasis and other Fungal Infections	10	0.6	0.8
resnet50	Urticaria Hives	10	0.6	0.7
resnet50	Vascular Tumors	10	0.5	0.8
resnet50	Vasculitis Photos	10	0.5	0.7
resnet50	Warts Molluscum and other Viral Infections	10	0.6	0.8
clip	Acne and Rosacea Photos	10	0.7	0.8
clip	Actinic Keratosis Basal Cell Carcinoma and other Malignant Lesions	10	0.9	1
clip	Atopic Dermatitis Photos	10	0.6	0.7
clip	Bullous Disease Photos	10	0.7	0.8
clip	Cellulitis Impetigo and other Bacterial Infections	10	0.5	0.5
clip	Eczema Photos	10	0.9	0.9
clip	Exanthems and Drug Eruptions	10	0.4	0.5
clip	Hair Loss Photos Alopecia and other Hair Diseases	10	0.6	0.8
clip	Herpes HPV and other STDs Photos	10	0.5	0.9
clip	Light Diseases and Disorders of Pigmentation	10	0.7	0.8
clip	Lupus and other Connective Tissue diseases	10	0.6	0.9
clip	Melanoma Skin Cancer Nevi and Moles	10	0.7	0.8
clip	Nail Fungus and other Nail Disease	10	0.7	0.8
clip	Poison Ivy Photos and other Contact Dermatitis	10	0.4	0.4
clip	Psoriasis pictures Lichen Planus and related diseases	10	0.6	0.8
clip	Scabies Lyme Disease and other Infestations and Bites	10	0.5	0.6
clip	Seborrheic Keratoses and other Benign Tumors	10	0.6	0.9
clip	Systemic Disease	10	0.4	0.6
clip	Tinea Ringworm Candidiasis and other Fungal Infections	10	0.7	1
clip	Urticaria Hives	10	0.8	0.9
clip	Vascular Tumors	10	0.6	0.7
clip	Vasculitis Photos	10	0.9	0.9
clip	Warts Molluscum and other Viral Infections	10	0.6	0.8

Retrieval Analysis

The nearest-neighbor query outcomes show that DINOv2 (**63.91%**) and CLIP (**63.48%**) strongly outperform ResNet50 (**53.91%**). This suggests that self-supervised representations can serve as highly effective backbones for content-based medical image retrieval (CBIR) systems, allowing clinicians to find visually similar cases from historical databases.

12. Discussion & Clinical Implications

The findings have significant implications for the clinical application of AI in dermatology:

- **Dataset Search & Curation:** DINOv2’s visual compactness makes it ideal for duplicate detection, outlier discovery, and data curation in large medical databases.
- **Annotation Support:** CLIP can be used to generate zero-shot candidate labels for unlabeled images, reducing clinician annotation loads.
- **Clinical CBIR:** Content-based image retrieval engines utilizing DINOv2 or CLIP can surface historical cases of morphologically similar lesions, acting as a clinical decision support tool.

13. Limitations

- **Class Imbalance:** The DermNet dataset contains a highly skewed class distribution, which biases metrics like Purity and Silhouette.
- **Visual Co-variables:** Factors such as patient skin tone, lesion body location, lighting, camera resolution, and the presence of hair add confounding features that models confuse with actual pathology.
- **Lack of Fine-Tuning:** Encoders were evaluated out-of-the-box; domain-specific fine-tuning would improve visual boundary resolution.

14. Conclusion

This study investigated the properties of pretrained visual representations for dermatological classification. We address our core research questions below:

1. **Does DINOv2 naturally separate dermatological diseases?** Only partially. While DINOv2 forms highly compact local structures, the visual clusters do not naturally separate into dermatological categories without domain-specific training. Instead, they group images by raw visual appearance (color, scale, texture).
2. **Which diseases remain difficult?** Diseases that present with overlapping morphological characteristics, such as erythematous scaly plaques (Psoriasis vs. Fungal Infections) and active localized inflammatory patches (Cellulitis vs. Contact Dermatitis), remain highly difficult to separate.
3. **Does CLIP outperform DINOv2?** CLIP outperforms DINOv2 on semantic alignment metrics (ARI, NMI, Purity) due to its contrastive text alignment, but DINOv2 produces more compact clusters (Silhouette) and slightly higher nearest-neighbor retrieval accuracy.
4. **Is clustering useful?** Unsupervised clustering is useful as a diagnostic probing tool to analyze representation geometry, but it is not sufficient as a standalone clinical diagnostic method due to low absolute alignment scores (ARI ~ 0.10).
5. **Which representation should future work build upon?** Future content-based image retrieval (CBIR) systems should build upon DINOv2 due to its visual retrieval accuracy (63.91%), while classification and indexing pipelines should utilize CLIP for semantic mapping.

15. Future Work

To translate these findings into clinically viable tools, future research should explore the following directions:

- **Domain-Specific Self-Supervised Learning (SSL):** Pretraining foundation models directly on large clinical skin lesion datasets (such as ISIC, HAM10000) using DINOv2 or contrastive learning objectives to teach the network medical features rather than general objects.
- **Multimodal Representation Fusion:** Integrating visual embeddings with clinical text metadata (patient history, lesion location, symptoms) using multimodal fusion layers to build a unified representation space.
- **Hierarchical Modeling:** Structuring the embedding space or loss functions to match the hierarchical diagnostic trees used by dermatologists, ensuring that visual confusions remain within parent diagnostic categories.
- **Prototype-Based Retrieval:** Learning clinical prototypes (exemplars) for each skin condition in the self-supervised space to assist clinicians with interpretability and differential diagnosis support.